Comparative Utilization & Architectural Analysis Report

Selective Hardening in RISC-V Soft-Processors for Safety-Critical & Radiation Environments

Rigorous cross-evaluation between reference paper **Cora et al. (Politecnico di Torino, Microelectronics Reliability 2025)** and the **RV32E Adaptive Fault-Tolerant Soft-Core Architecture (Project: Final-Project-EV)**.

Reference: Cora et al. (ESREF 2024 / MR 2025) | Project: RV32E Fault-Tolerant EV Core | AMD Xilinx Zynq-7000 & Artix-7 | Cadence 90nm/180nm ASIC Flow | ISO 26262 ASIL-D

1. Executive Summary & Research Scope

This utilization report presents a quantitative, dual-sided comparative evaluation between the reference publication "*Selective hardening of RISCV soft-processors for space applications*" by **G. Cora, C. De Sio, S. Azimi, and L. Sterpone (Politecnico di Torino, 2025)** and our **RV32E Fault-Tolerant Soft-Core Architecture** (*Final-Project-EV*).

The reference paper investigates the vulnerability profile of the **NEORV32** soft-processor implemented on an AMD Xilinx Zynq-7020 SoC (XC7Z020) under radiation-induced Single Event Upsets (SEUs) and Multiple Bit Upsets (MBUs). Its central finding proves that applying monolithic Triple Modular Redundancy (TMR) across an entire processor results in an unacceptable **300% overhead penalty** across logic, registers, and Block RAM. Instead, *selective hardening* applied to the most critical modules (the Control Unit and ALU) achieves near-identical fault mitigation while drastically reducing silicon resource consumption.

Our project adopts and extends this selective hardening paradigm, tailoring it to the rigorous demands of **Electric Vehicle (EV) powertrain control and ISO 26262 ASIL-D functional safety**. Rather than relying on blunt 3-way replication across the board, our architecture introduces a **heterogeneous, energy-proportional fault-tolerance framework**:

- **Adaptive Redundancy Controller (ARC) on ALU:** Dynamically toggles between low-power Simplex execution and high-reliability TMR majority voting with operand isolation and glitch-free clock gating.
- **Dual Modular Redundancy (DMR) with Fail-Safe Clamping on Control Unit:** Detects 100% of decoder anomalies via XOR shadow comparison and instantly clamps destructive write signals (`safe_mem_write` , `safe_reg_write`) to zero, avoiding voter gate delays on 10+ control wires while saving 33% area compared to CU TMR.
- **Triple Modular Redundancy (TMR) on Program Counter:** 3-way replicated PC register bank with majority voting to guarantee continuous instruction sequencing.
- **Extended Hamming(38,32) SEC-DED ECC on Register File:** Single-bit correction and double-bit error detection with only 21.8% bit overhead, avoiding the 200% storage overhead of register triplication.
- **Native Built-In ECC on Block RAM:** Utilizing Xilinx `RAMB36E1` dedicated hardware primitives, completely eliminating the 200% BRAM memory explosion reported in the paper.

2. System Architecture & Baseline Specifications

ARCHITECTURAL PARAMETER	REFERENCE PAPER: CORA ET AL. (2025)	OUR PROJECT: FINAL-PROJECT-EV
Core Architecture	**NEORV32** (S. Nolting et al.) — Multi-cycle / Pipelined general-purpose soft-processor.	**RV32E Custom Core** — Deterministic 3-stage pipeline (IF → ID/EX → WB) optimized for bounded WCET.
Instruction Set (ISA)	RV32I [C, M, Zfinx] + Zicntr, Zihpm (Integer, Compressed, Multiply/Divide, Floating-Point).	RV32E Base Integer Profile (32-bit embedded profile with 16 registers: `x0` – `x15`).
Register Count	32 architectural registers (32-bit width).	16 architectural registers (`x0` – `x15`, 32-bit width) with 39-bit ECC storage.
Target Hardware Fabric	AMD Xilinx Zynq-7020 SoC (XC7Z020-1CLG400C) on PYNQ-Z2 board.	• AMD Xilinx Artix-7 (XC7A100T-1CSG324C) on Nexys 4 • AMD Xilinx Zynq-7010 (XC7Z010-1CLG400C) on Zybo Z7 • Cadence Genus / Innovus 90nm/180nm ASIC Flow
Operating Frequency	Standard FPGA soft-core (~25.0 – 50.0 MHz).	**25.0 MHz** (Period: 40.0 ns, WNS = +10.668 ns, F_max ≈ 34.1 MHz).
Application Domain	**Aerospace / Deep Space Radiation** (PSI proton beam testing model).	**Electric Vehicle (EV) Powertrain & Embedded Functional Safety** (ISO 26262 ASIL-D).
Failure Mode Priority	**Fail-Operational:** System must keep executing through particle strikes.	**Fail-Safe Containment:** Destructive writes clamped; instant deterministic trap.

3. Master Hardware Utilization Benchmark Matrix

The table below presents the quantitative resource utilization comparison across the five NEORV32 configurations documented in Table 1 of Cora et al. (Zynq-7020 fabric: 53,200 LUTs, 106,400 FFs, 140 BRAM36 tiles, 220 DSPs) and our implemented designs on the Artix-7 XC7A100T (63,400 LUTs, 126,800 FFs, 135 BRAM36 tiles, 240 DSPs) and Zynq-7010 SoC (17,600 LUTs, 35,200 FFs, 60 BRAM36 tiles, 80 DSPs).

DESIGN IMPLEMENTATION	SLICE LUTS	LUT UTIL	SLICE REGISTERS (FF)	FF UTIL	BRAM TILES (36KB EQ.)	BRAM UTIL	DSP BLOCKS	DSP UTIL
Paper: NEORV32 Plain (Baseline)	3,203	6.02%	1,990	1.87%	13.5	9.64%	2	0.91%
Paper: NEORV32 Full TMR	9,672	18.18%	5,980	5.62%	40.5	28.93%	6	2.73%
Paper: NEORV32 ALU-CU TMR	7,570	14.23%	4,990	4.69%	13.5	9.64%	6	2.73%
Paper: NEORV32 CU-BUS TMR	5,969	11.22%	3,575	3.36%	13.5	9.64%	2	0.91%
Paper: NEORV32 RF-BUS TMR	3,203	6.02%	2,149	2.02%	16.5	11.79%	2	0.91%
Our Core: RV32E Core (Artix-7 XC7A100T)	**3,807**	**6.00%**	**788**	**0.62%**	**0.0** (OOC)	**0.00%**	**0**	**0.00%**
Our SoC: Complete Zybo Z7 (XC7Z010)	**~4,450**	**25.28%**	**~1,820**	**5.17%**	**2.0**	3.33%	**0**	**0.00%**

> **Key Benchmark Finding: Preventing the 300% BRAM Memory Explosion**
> In Cora et al., applying Full TMR caused Block RAM consumption to skyrocket from **9.64% to 28.93%** (consuming 40.5 BRAM tiles). Our architecture solves this completely: standalone core synthesis consumes **0 BRAM**, while the complete SoC instantiates 32 KB / 4 KB unified memory with **built-in hardware SEC-DED ECC (RAMB36E1)**, requiring only **2 tiles (3.33% of Zybo Z7)** and incurring **0% memory replication overhead**.

4. Deep-Dive Comparative Factor Analysis

Factor 1: Control Unit (CU) Hardening — Fail-Safe Clamping vs. 3-Way Voting

Table 2 of the reference paper establishes that the Control Unit is the single most vulnerable entity in the processor, containing **238,327 Essential Bits (26.48% of the core total)**. In the paper, triplicating the CU increased LUT utilization from 6.02% to 14.23%. Furthermore, voting across 10+ control wires (`reg_write`, `mem_write`, `alu_op[3:0]`, `wb_sel[1:0]`, etc.) introduces severe critical-path gate delays and routing congestion.

Cora et al. Approach: Full CU TMR

- Instantiates 3 complete Control Unit decoders.
- Requires 3-way majority voters on every decoded wire.
- Voter gate propagation delays worsen overall clock slack.
- Consumes +8.21% additional LUTs across the FPGA fabric.

Our Approach: DMR with Fail-Safe Clamping

- Instantiates 2 decoders (Primary + DMR Checker).
- Single XOR tree detects any discrepancy between outputs.
- Instantly clamps destructive signals (`mem_write & ~cu_mismatch`).
- **Saves 33% area** compared to CU TMR; zero voter delay.

Factor 2: ALU Protection & Dynamic Energy Proportionality

Cora et al. demonstrated that ALU errors dominate floating-point workloads (reaching a 26.55% MBU error peak) but drop below 4% under integer workloads. However, their ALU triplication remains permanently active, continually dissipating dynamic switching power regardless of the workload phase.

Our architecture introduces the **Adaptive Redundancy Controller (ARC)** (`adaptive_redundancy_controller.sv`):

- **Hardware Pin / MMIO Mode Control:** Real-time toggling between Simplex (Mode 0) and TMR (Mode 1).
- **Glitch-Free Integrated Clock Gating (ICG):** Driven by active-low falling-edge latches (`clock_gater.sv`), cutting clocks to redundant execution units `u_alu_1` and `u_alu_2` during Simplex execution.
- **Operand Clamping:** In Simplex mode, operands to redundant ALUs are clamped to `32'h0` and opcodes to neutral `ALU_ADD`, eliminating dynamic logic toggling and saving up to **66% ALU dynamic execution power**.

Factor 3: Register File (RF) Protection — SEC-DED vs. Triplication

In Table 1 of Cora et al., the RF-BUS TMR configuration yielded negligible reliability benefits because triplicating 32 registers tripled storage flip-flops without addressing the vulnerability of the ALU or CU.

Our project deploys **Extended Hamming(38,32) SEC-DED ECC** on the 16-entry register file (`regfile.sv`):

- Stores 39 bits per register (32 data bits + 6 Hamming parity bits + 1 overall parity bit).
- **Single Error Correction (SEC):** Automatically calculates syndrome bits and flips corrupted bits on-the-fly with **zero pipeline stalls**.
- **Double Error Detection (DED):** Detects uncorrectable double-bit flips and asserts telemetry flags (`ecc_ded_1` , `ecc_ded_2`).
- **Storage Efficiency:** Incurs only **21.8% bit overhead** (7 check bits / 32 data bits) compared to the **200% storage overhead** of register triplication.

5. Detailed FPGA Primitives Inventory & Physical Implementation

PRIMITIVE CATEGORY	PRIMITIVE REF NAME	USED (CORE TOP)	AVAILABLE (XC7A100T)	UTIL (%)	MICROARCHITECTURAL FUNCTION & PURPOSE
Logic LUT	`LUT6`	2,532	63,400	3.99%	Wide instruction decoding, SEC-DED syndrome XOR trees, 32-bit majority voters.
Logic LUT	`LUT5`	516	63,400	0.81%	Dual-output slice packing, forwarding multiplexers, branch target generators.
Logic LUT	`LUT2 / LUT3 / LUT4`	1,319	63,400	2.08%	Control signal clamping (`safe_mem_write`), immediate generation, clock enables.
Carry Logic	`CARRY4`	131	15,850	0.83%	Fast carry arithmetic for triplicated 32-bit ALUs and TMR PC incrementers (`pc + 4`).
Wide Multiplexers	`MUXF7 / MUXF8`	125 / 5	31,700 / 15,850	0.39% / 0.03%	Asynchronous read muxes for the 16-entry register file dual read ports (Port 1 & Port 2).
Flip-Flops	`FDCE`	785	126,800	0.62%	Pipeline registers (IF/ID, ID/EX, EX/WB), 16x39-bit ECC register array, TMR PC bank.
Flip-Flops	`FDPE`	3	126,800	<0.01%	Asynchronous preset registers for mode registers and power-on reset sequencing.
DSP Multipliers	`DSP48E1`	**0**	240	**0.00%**	Zero DSP primitives required. Multiplier-free RV32E keeps fabric pure combinational.
Block Memory	`RAMB36E1 / RAMB18E1`	**0** (Core) / **2** (SoC)	135 / 270	**0.00% / 1.48%**	Core is memory-agnostic; SoC instantiates unified 32 KB BRAM with hardware SEC-DED.

6. Timing, Power, and Electrical Analysis

ELECTRICAL / TIMING METRIC	REPORTED VALUE (POST-ROUTE)	TIMING CONSTRAINT / OPERATING SPEC	MARGIN / COMPLIANCE STATUS
Clock Frequency	**25.000 MHz**	40.000 ns Period	PASSED (Target Met)
Worst Negative Slack (WNS)	**+10.668 ns**	> 0.000 ns	PASSED (29.332 ns Critical Path → F_max ≈ 34.1 MHz)
Worst Hold Slack (WHS)	**+0.181 ns**	> 0.000 ns	PASSED (Zero hold violations across 1,813 endpoints)
Total On-Chip Power	**0.111 W (111 mW)**	Device Thermal Budget: 1.500 W	EXCELLENT (92.6% thermal headroom remaining)
Dynamic Power Dissipation	**0.027 W (27 mW)**	24.3% of Total Power	Dominated by register file SEC-DED syndrome calculations (24 mW).
Device Static (Leakage) Power	**0.084 W (84 mW)**	75.7% of Total Power	Baseline quiescent leakage of 28nm Artix-7 silicon substrate.
Junction Temperature (T_j)	**25.5 °C**	Max Operating: 85.0 °C	Near ambient (T_amb = 25.0 °C); minimal internal heat generation.

7. Fault Vulnerability Analysis: Essential Bits Cross-Comparison

Cora et al. performed exhaustive fault injection across the Configuration RAM (CRAM) using AMD SEM-IP and the PyXEL toolkit. Their results quantified the *Essential Bits* (CRAM bits that directly alter circuit behavior if hit by a particle) for each sub-module:

PROCESSOR SUB-MODULE	ESSENTIAL BITS (CORA ET AL.)	% OF CORE ESSENTIAL BITS	MITIGATION IN CORA ET AL.	MITIGATION IN OUR PROJECT (FINAL-PROJECT-EV)
Control Unit (CU)	**238,327**	**26.48%**	Full 3-Way TMR (10+ voters)	**DMR + XOR Tree + Fail-Safe Clamping + Deterministic Trap**
Arithmetic Unit (ALU)	**183,539**	**20.39%**	Static 3-Way TMR	**Adaptive Redundancy Controller (ARC) + Glitch-Free Clock Gating**
Internal Memories (MEMs)	**43,122**	**4.79%**	Triplication in Full TMR	**In-Tile Hardware SEC-DED ECC (RAMB36E1 / ASIC SEC-DED Wrapper)**
Register File (RF)	**27,021**	**3.00%**	Triplication in RF-BUS TMR	**Extended Hamming(38,32) SEC-DED (39-bit storage, zero stalls)**
Internal Bus (BUS)	**17,730**	**1.97%**	Triplication in CU-BUS TMR	**Tightly-coupled zero-cycle register forwarding & hazard interlocks**
Program Counter (PC)	*(Lumped in IF)*	*(Lumped in IF)*	Unprotected in selective modes	**Dedicated 3-Way TMR PC Register Bank (`pc_reg_a/b/c`) + Voter**

8. Conclusions & Strategic Engineering Takeaways

Conclusion 1: Validation of Selective Hardening

Both the reference publication and our experimental results confirm that **monolithic TMR is an inefficient engineering choice**. Triplicating an entire processor triples logic, registers, and memory without providing proportional reliability benefits over selective protection.

Conclusion 2: The Control Unit is the Core Linchpin

Cora et al. demonstrated that the Control Unit accounts for over 26% of all critical bits. Protecting only the datapath (ALU) while leaving the Control Unit vulnerable is dangerous. Our project's **Dual Modular Redundancy (DMR) with fail-safe clamping** achieves 100% write-corruption prevention while saving 33% silicon area compared to CU TMR and eliminating multi-signal voter propagation delays.

Conclusion 3: Memory Demands In-Tile ECC, Not Triplication

Cora et al.'s Full TMR design tripled Block RAM utilization from 9.64% to 28.93%. Our project demonstrates that utilizing dedicated **hardware SEC-DED ECC (RAMB36E1)** provides superior Single Event Upset correction with **zero tile duplication overhead**.

Conclusion 4: Dynamic Energy Proportionality (ARC)

Unlike the static redundancy of the reference paper, our **Adaptive Redundancy Controller** with glitch-free clock gating allows runtime switching between Simplex (low-power cruising) and TMR (safety-critical braking/steering) modes, cutting ALU dynamic power by 66% when full redundancy is unnecessary.